

Assessment 3

N+ Network security

Section A: Scenario-Based Concept Application

1. Explain the selection process for CPU, RAM, and storage specifications to ensure hardware compatibility and prevent performance bottlenecks for a startup team performing concurrent development and video conferencing tasks.

❖ When building workstations for a startup team handling:

- Software development
- Multiple browser tabs
- Virtual machines
- Video conferencing
- Collaboration tools

➤ the hardware must avoid performance bottlenecks and maintain compatibility.

❖ **CPU Selection**

➤ The CPU handles calculations, multitasking, and application execution.

• **Important Factors**

- Core count
- Thread count
- Clock speed
- Cache memory
- Motherboard socket compatibility

• **Recommended Choice**

- Minimum: 6-core CPU
- Recommended: 8-core or higher

❖ **RAM Selection**

➤ RAM temporarily stores active applications and processes.

• **Considerations**

- Capacity
- Speed
- DDR generation compatibility
- Dual-channel support

• **Recommended Capacity**

- Minimum: 16 GB
- Recommended: 32 GB

❖ **Storage Selection**

- Storage affects:
 - Boot speed

- Application loading
- File access performance
- **Recommended Storage**
 - NVMe SSD preferred
 - Minimum 512 GB

❖ **Compatibility Considerations**

- Ensure:
 - CPU socket matches motherboard
 - RAM type supported by motherboard
 - PSU wattage supports all components
 - Cooling system fits cabinet size
- Proper hardware planning prevents:
 - Thermal throttling
 - Slow performance
 - Hardware instability

2. Explain how system resource allocation and hardware bottlenecks contribute to performance degradation after the deployment of resource-intensive background applications like antivirus software.

- Background applications like antivirus software continuously consume:
 - CPU cycles
 - RAM
 - Disk I/O
 - Network bandwidth
- When resource-intensive antivirus scans begin:
 - CPU usage spikes
 - RAM becomes heavily utilized
 - Disk access increases

❖ **Example Scenario**

- A startup workstation already runs:
 - IDE
 - Browser
 - Teams/Zoom
 - Virtual machine

❖ **Performance Impact**

- Users may experience:
 - Freezing
 - Delayed response
 - Lag during meetings

- Slow application launches
 - Efficient resource allocation and scheduled background scans help minimize degradation.
3. Explain the systematic troubleshooting process for resolving peripheral hardware failures caused by driver incompatibility or operating system conflicts when a device functions inconsistently across different workstations.
- When a peripheral works inconsistently across workstations, the issue often relates to:
 - Drivers
 - OS compatibility
 - USB controller conflicts
 - Firmware issues

❖ **Systematic Troubleshooting Process**

Step 1: Verify Physical Hardware

Check:

- USB cable
- Power supply
- Port damage

Step 2: Test on Another Workstation

If the device works elsewhere:

- Hardware is likely functional
- Problem is system-specific

Step 3: Check Device Manager

Look for:

- Yellow warning icons
- Unknown devices
- Driver errors

Step 4: Update/Reinstall Drivers

Install:

- Manufacturer-certified drivers
- Correct OS version drivers

Step 5: Check Operating System Conflicts

Possible issues:

- Outdated OS patches
- Conflicting software
- Disabled services

Step 6: Review Event Logs

Check:

- Driver crash logs
- USB controller errors
- Kernel failures

➤ The troubleshooting process isolates whether the problem originates from:

- Hardware
- Driver
- OS configuration
- Software conflict

4. Explain the role of Uninterruptible Power Supply (UPS) systems and surge protection in preventing file system corruption and ensuring data integrity during unexpected power fluctuations.

➤ **UPS (Uninterruptible Power Supply)**

- A UPS provides temporary battery power during:
 - Power outages
 - Voltage drops
 - Power instability

❖ Importance of UPS

Prevents Sudden Shutdowns

- Corrupt files
- Damage databases
- Interrupt OS operations

❖ Protects File Systems

➤ Modern operating systems constantly write data to disk.

Unexpected shutdowns can corrupt:

- NTFS/ext4 file systems
- Boot sectors
- Application data

❖ Ensures Data Integrity

UPS systems allow:

- Proper shutdown procedures
- Save operations
- Backup completion

❖ Combined Benefit

UPS + surge protection improve:

- Hardware lifespan
- Data reliability
- Business continuity

Section B: Practical Application

1. Physically assemble a workstation and document the component installation sequence.

Component Installation Order

1. Install CPU on motherboard
2. Apply thermal paste
3. Install CPU cooler
4. Insert RAM modules
5. Mount motherboard into case
6. Install Power Supply Unit (PSU)
7. Install SSD/HDD storage
8. Connect SATA and power cables
9. Install GPU (if required)
10. Connect front panel connectors
11. Connect case fans
12. Verify all power connections
13. Power on system and enter BIOS

2. Troubleshoot and resolve a RAM mismatch error within the BIOS/UEFI settings.

❖ Symptoms

- BIOS memory warning
- Random crashes
- System instability

❖ Troubleshooting Steps

Step 1 Enter BIOS/UEFI setup.

Step 2 Verify:

- RAM frequency
- Voltage
- Capacity recognition

Step 3 Disable XMP temporarily.

Step 4 Ensure:

- Matching RAM generations
- Correct DIMM slot placement

Step 5 Update BIOS firmware if necessary.

❖ **Resolution**

Set RAM to:

- Supported motherboard speed
- Stable timings

❖ **Result:**

- Successful POST
- Stable operation

3. Build a LAN topology in Packet Tracer for 5 systems and 1 printer using a Class C subnet.

❖ **Devices**

- 5 PCs
- 1 Printer
- 1 Switch

❖ **Class C Subnet**

- Network: 192.168.1.0/24

IP Assignment Table

Device	IP Address
PC1	192.168.1.10
PC2	192.168.1.11
PC3	192.168.1.12
PC4	192.168.1.13
PC5	192.168.1.14
Printer	192.168.1.20

- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1

4. Perform a clean OS installation with manual disk partitioning and startup optimization.
Success Criteria: System passes POST, all IPs are reachable via Ping, and the OS boots without errors.

❖ **Installation Steps**

Step 1 Boot using:

- Windows/Linux ISO
- USB installer

Step 2 Choose:

- Custom installation

Step 3 Create manual partitions.

Example:

Partition	Size
System	100 GB
Applications	200 GB
Data	Remaining

Step 4 Format partitions:

- NTFS (Windows)
- ext4 (Linux)

Step 5 Install OS and drivers.

Startup Optimization

➤ Disable unnecessary startup applications:

- Auto-updaters
- Unused background apps
- Excess startup services

Use:

- Task Manager (Windows)
- systemctl (Linux)

Section C: Integrated Case Study

Scenario: At Nexus Innovations, three rendering workstations pass morning POST checks but restart randomly during heavy afternoon workloads. The units are stored in poorly ventilated desk cabinets.

1. Identify the three hardware components most likely to cause random reboots under load.

❖ **CPU**

➤ Heavy rendering generates high heat.

❖ **PSU (Power Supply Unit)**

➤ Insufficient or failing power delivery under load.

❖ **RAM**

➤ Faulty or overheating memory modules may trigger crashes.

❖ Additional possibility:

➤ GPU overheating during rendering

2. Explain why the systems pass POST in the morning but fail in the afternoon.

❖ Morning conditions:

- Cooler room temperature
- Lower internal cabinet heat

❖ Afternoon conditions:

- Poor ventilation traps heat
- Continuous rendering increases thermal load

❖ As temperature rises:

- CPU thermal protection activates
- PSU efficiency drops
- Components become unstable

❖ POST succeeds because:

- Initial startup occurs before temperatures rise significantly.

3. How would you use a stress-test tool to confirm a thermal or power issue?

❖ **Tools**

Examples:

- Prime95
- AIDA64
- FurMark
- OCCT

Step 1 Monitor temperatures using:

- HWMonitor
- Open Hardware Monitor

Step 2 Run CPU/GPU stress tests.

Step 3 Observe:

- Temperature spikes
- Voltage drops
- Sudden shutdowns
- Reboots

❖ **Confirmation**

- If reboot occurs under stress:
- Thermal issue or PSU instability is confirmed.

4. Propose three specific steps to remediate the frequent system restarts.

❖ **Improve Airflow**

- Remove systems from enclosed cabinets
- Add ventilation space

❖ **Clean Cooling Systems**

- Remove dust buildup
- Replace clogged filters

❖ **Upgrade Cooling**

- Install better case fans
- Reapply thermal paste
- Upgrade CPU cooler

❖ Additional step:

- Maintain cooler ambient room temperature

5. Provide a brief technical justification for why a PSU upgrade might (or might not) be necessary.

❖ **Why Upgrade May Be Necessary**

- A weak PSU may:
 - Fail under peak rendering loads
 - Deliver unstable voltage
 - Trigger sudden shutdowns
- Especially if:
 - GPU power demand increased
 - PSU wattage is insufficient

❖ **Why Upgrade Might NOT Be Necessary**

- If testing shows:
 - Stable voltage rails
 - Adequate wattage
 - No shutdown during isolated PSU testing
- then the primary issue is likely:
 - Overheating
 - Poor ventilation
- Therefore:
 - Thermal diagnostics should be completed before replacing the PSU.